Multi-Algorithm Comparison Report

Common benchmark using the same input, calibration and shared evaluation metrics.

Input file	1.png
Algorithms compared	4
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Decision Summary

Recommended algorithm: **Unsharp Masking**. The recommendation uses the transparent balanced quality score below; it does not claim biometric matching accuracy or substitute for clean-reference validation.

Component	Weight	Normalisation
Contrast	15%	clip(CII / 1.5, 0, 1)
Ridge-valley clarity	25%	clip(output/input ratio / 1.5, 0, 1)
Edge clarity	20%	clip(output/input ratio / 1.5, 0, 1)
Structural preservation	40%	SSIM clipped to 0-1

Visual Comparison

Original

RHLT

Gabor Filter Bank

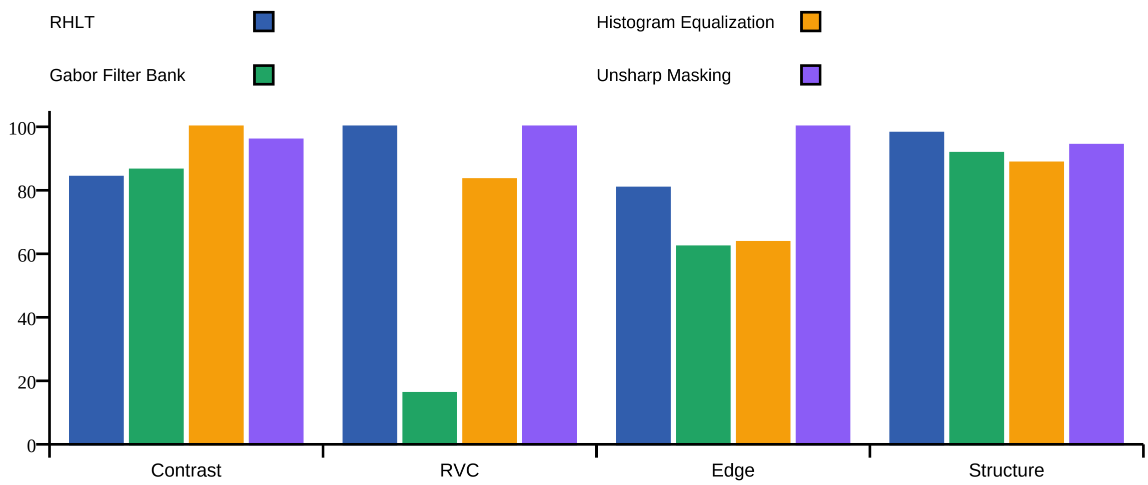
Histogram Equalization

Unsharp Masking - Recommended

Quantitative Benchmark

Algorithm	Score	CII	Sharp. delta	Edge delta	SSIM	RVC delta	Time (ms)
RHLT	93.0/100	1.26x	+56.8%	+21.1%	0.980	+58.2%	1912.1
Gabor Filter Bank	66.1/100	1.30x	-72.3%	-6.7%	0.917	-75.9%	375.9
Histogram Equalization	84.0/100	1.72x	+25.6%	-4.6%	0.886	+25.1%	2228.2
Unsharp Masking	97.1/100	1.44x	+720.6%	+66.0%	0.942	+739.9%	415.5

Normalised Quality Components (Higher Is Better)



SSIM measures structural preservation against the original 1.000 reference; it is not reported as an enhancement percentage. Runtime is excluded from the quality score and must be considered separately.

Best Algorithm Awards

Award	Algorithm	Detail
Best balanced quality	Unsharp Masking	97.1/100
Fastest processing	Gabor Filter Bank	375.9 ms
Highest sharpness	Unsharp Masking	+720.6%